
OMR1100



CHATSWORTH DATA
CORPORATION

SECTION 1 SERVICE AND TECHNICAL SUPPORT

In the event that your OMR-1100 requires service or repair, Chatsworth Data Corporation provides the following options:

- Standard Repair: OMR is repaired and reshipped within 7 working days.
- Priority Repair: OMR is repaired and reshipped within 24 Hours.
- Exchange: Reader is replaced with another (when available) within 24 hours.

All readers in need of service or repair require an RMA Number (Return Merchandise Authorization).

When contacting our Customer Service Department for the RMA Number, please provide the model and serial numbers. From the I.D. label located on the bottom of the OMR.

When shipping the OMR back to Chatsworth Data Corporation reference the RMA number on the outside shipping container or the shipping label. This will speed up the repair process and insure control over the return of the unit.

If possible, ship the reader back in the original container in which your OMR-1100 was received. If the original container is not available, please provide adequate packing, to insure that the unit will not be damaged in transit.

Ship To:

Chatsworth Data Corporation
20710 Lassen Street
Chatsworth, Ca. 91311
Ref. RMA# "X-XXXXXXX"
(Where X-XXXXXXX" is the RMA Number received from customer Service)

Customer Service And Technical Support:

818-341-9200
800-423-5217 (Outside Ca.)
818-341-3002 (Fax)

SECTION 2 GENERAL INFORMATION

ABOUT THE MANUAL

This manual provides instructions for installing, testing and programming for the OMR-1100, manufactured by the Chatsworth Data Corporation.

The manual is divided into sections to assist you in referencing the information desired. Each Section has its title at the top of the page. Each section is divided into topics such as "About The Manual" to further direct you to specific topics. Specific topics headings are outlined within each general topic and are in Upper case.

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STANDARD CONVENTIONS

Specific requirements or actions are IN ALL CAPS and are indicated by the word NOTE: in an indented paragraph.

Example:

NOTE: DO NOT USE ANY SERIAL CABLE OTHER THAN THE ONE SUPPLIED WITH THE OMR.

Data to be entered via a keyboard will be displayed in [] symbols. The symbols are not part of the input. In the following example you would be asked to type OMRDEMO and then press the Enter Key:

Example: C:\>[OMRDEMO] [Enter]

Information displayed at the DOS > is shown based on using DOS 6.2. If you are using other than DOS 6.2 your display may vary slightly.

In the following example you would be asked to type CD\Utilities at the DOS prompt:

Example: C:\>[CD\Utility][Enter] C:\UTILITY>

Syntax variables that are used when communicating with the OMR are displayed in Microsoft QB4.5 format.

All source code examples are shown using Microsoft QB (Quick Basic) version 4.5

ABOUT THE OMR-1100

The OMR-1100 is a programmable, microprocessor-based Optical Mark Reader. The OMR is designed to scan a standard 12 channel, 3.25in. wide response document and convert the marked data to an ASCII character format that can be transmitted to a computer via a serial data communications port. (RS232C).

The conversion of the response marks to data may utilize any of four (4) internal formats generated by the OMR logic.

They are:

- Hollerith-To-ASCII Conversion
- Binary-To-ASCII Conversion
- Translate Conversion
- Split Translate Conversion

A more complete description of each of the Conversion modes and how they may be utilized is discussed in Section 5.

RESPONSE DOCUMENTS

There are numerous and varied ways to design a document to be used on the OMR-1100. However, a basic card utilizes a maximum of 12 data channels (defined as 12 rows which run the length of the document).

There are up to 126 vertical columns that may be dispersed over the length of the document. There is no specific maximum length of the document, but practical considerations normally limit the document length to between 7 3/8" and 14". A minimum of 3" is required to transport the document.

Each vertical column is defined with a timing mark along the left edge of the card to define the portion of the response area that will be scanned for the presence of a mark.

Several card samples are included with your OMR and a more complete discussion of document design, tips, suggestions and specifications are included in Appendix A.

SERIAL COMMUNICATIONS:

The OMR-1100 utilizes a standard serial communication mode defined as RS232/C. This communication protocol will support baud rates from 300 to 38400 on any installed communication port.

The data string format is defined as One Start Bit, 7 Data Bits, Even Parity and One Stop Bit.

The baud rate will default to 9600 when the OMR is powered up. The default baud rate is adequate for nearly all applications and computers. However, if necessary, the baud rate may be modified after power up utilizing the utility program OMRCfg.EXE. Complete instructions for executing OMRCFG.EXE are included in Section 8.

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SECTION 3: INSTALLATION And CHECKOUT

INSTALLATION

Before using the OMR1100 to enter data, proceed as follows:

Plug the three-prong end of the power cord from the transformer into a 110 VAC outlet. The other end of the power cord from the transformer should have a round male end that is to be inserted into the power receptacle on the right side of the reader.

Next, attach the data cable from the OMR1100 to the computer. The male end of the data cable will attach to the 25 pin female connector on the right side of the OMR. Attach the female end of your data cable to the communications port on the computer.

If there is a serial "mouse" attached to your computer it will most likely be using the COM1 port. The remaining serial ports should be COM2, COM3, Etc. If you are not sure which communications port the OMR1100 is attached to, it can be identified by using the Utility Program OMRDetect discussed in the CHECKOUT section.

NOTE: There are 3 different data cables available from Chatsworth Data.:

PC cable uses:	25pin male and female connectors.
AT cable uses:	25pin male and 9 pin female connectors.
Macintosh cable uses:	25 pin male and 9 pin DIN female connectors.

If you did not receive the correct cable with your OMR1100, contact Customer Service for a replacement cable. DO NOT attempt to use a standard serial cable. The cable supplied by Chatsworth Data is custom made specifically for use with the OMR1100.

The On/Off switch is also located on the right side of the OMR. Press down on the side of the switch with the small white dot to apply power to the OMR1100.

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CHECKOUT

To insure the unit is powered up take one of the sample response forms supplied and insert it into the throat of the OMR1100. The motors should turn on and transport the card through the OMR. If there is no response from the OMR, check the power hookups making sure all connections are secure and that the AC receptacle you are using has power.

If The Unit still does not respond, contact Chatsworth Data Customer Service for assistance.

This is a good time to utilize the utility programs to check out the installation. The first Utility Program to use is OMRDetct.EXE.

Assuming that the Utility Programs have NOT been installed on the hard drive, place the Utility Distribution Disk in drive A or B and proceed as follows:

At the C:\> Type [A:] or [B:] and Press [Enter]
At the A:\> type [OMRDetct.EXE].
Press [Enter] key

This Utility program will examine the System configuration to which the OMR1100 is attached to determine which Communications port is being used. It will also determine the current baud rate of the OMR and, if possible, determine that the proper data cable is being used. This information will be displayed on the screen after the process is complete.

If the program is unable to communicate with the OMR, appropriate messages will display. If this should happen, recheck the data cable connections, make sure the reader has power and try again. If the problem still exists, contact Chatsworth Data Customer Service.

Assuming that the OMRDetct program executed satisfactorily, the Utility program called OMRDemo will confirm that the OMR1100 is reading properly.

Find the sample response form used to make sure the unit was powered up, and using a pencil make several marks in the response "bubbles". Since the next Utility program will display an image of the marks on a card, it helps if you make your marks in a pattern. i.e. mark the A in question 1, the B in question 2, the C in question 3, etc.

At the A:\> type [OMRDemo].
Press [Enter] key

The screen will clear and a graph form will display with the rows of the card displayed vertically and the columns of the card displayed horizontally. You will then be prompted to insert a card.

Insert the card you just marked into the OMR1100. As soon as the card has been transported, an image of the marks will display.

An example of a typical display is shown in Figure-1.

Sample Demo Screen:

Figure - 1

Chatsworth OMR1100

The image displayed should be reflective of the marks made on the response form. If some of the marks do not display, examine the marks to make sure that they completely fill the mark area and that there is no unmarked areas within the mark. This is especially important if the marking area is an oval and the mark was made with a circular motion leaving the centre of the oval clear. Correct any mis-marked areas and re-scan the document. If there still appears to be problems, contact Chatsworth Data Customer Service. Upon successful completion of the checkout utilities the OMR1100 is ready to fulfil the document scanning requirements of your application program.

Note: If the Document Feed Control option was ordered for the OMR, a document will not transport unless the OMR is under the control of an application computer program. Consequently, you will not be able to run the checkout procedures as defined in this Section.

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SECTION 4 DISTRIBUTION SOFTWARE

CHATSCORE

Chatscore was developed by Chatsworth Data Corporation as a stand-alone Teacher Made Test Scoring and Survey package for use by educational institutions and industry.

This application program supports the scoring of multiple choice or true and false tests developed by a teacher or as part of a specific testing instrument. Over 25 different test scoring card formats are supported by the program.

The system supports entering the correct answers, processing the student response forms and prints a series of reports. Chatscore also contains a survey module that will support up to 100 multiple choice responses and produces a summarized item analysis for each question.

A complete operator's guide for Chatscore can be printed or viewed by running the CSManual.EXE program included on the distribution disk.

To install the Chatscore programs on your hard drive, create a subdirectory of your choice (CHATSCOR is a good directory name) and install the system using the Install.Exe program as follows:

```
C:\>[MD\CHATSCOR] [Enter]
C:\>[CD\CHATSCOR] [Enter]
C:\CHATSCOR>[INSTALL.EXE] [Enter]
```

The Chatscore system is comprised of the following programs and files;

CSManual.exe: Print or Display Chatscore Manual

Install.EXE: Install Chatscore to hard drive

Setup5.Exe: Setup Program

Setup5.DES: Configuration File

Brun45.EXE: QB Run-time module

CS_Exel2.EXE: Excelsior Grade Export

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OMR-1100 UTILITY PROGRAMS

A series of utility programs have been supplied on your distribution disk. These programs will assist you in the setup and checkout of your OMR-1100 and others may be helpful in utilizing some of the advanced features of your OMR-1100.

To install the OMR utility programs on your hard drive, create a subdirectory of your choice (OMRUtil is a good directory name) and copy the utility program as follows:

```
C:\>[MD\OMRUtil] [Enter]
C:\>[CD\OMRUtil] [Enter]
C:\OMRUtil>[Copy A:OMR*. * C:] [Enter]
```

It is highly recommended that the OMR utility programs be installed on any computer that may be using an OMR-1100 to facilitate technical support.

The Utility programs are included in the following files;

OMRCfg.EXE:	Configuration Program
OMRCfg.CFG:	Configuration File
OMRDemo.EXE:	Card Image Display
OMRDetct.EXE:	Detect OMR Com: Port, including baud rate and cable information.
OMRDnld.EXE:	OMR Universal Driver

For complete instructions related to utilizing the Utility programs please reference Section 8.

If you are going to be using the OMR-1100 to support an application program developed for use with an OMR-1100, there is no need to continue reviewing the remaining sections of this manual, as they are for Programmers and Systems Analysts.

However, it is recommended that the operations manual of the application program be reviewed to determine if it is necessary to specify the COM port and the Baud rate of the OMR within the application program.

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SECTION 5: DATA MARK CONVERSION

The Translation Download feature of the OMR-1100 allows a series of commands to be sent to the OMR via the Serial Communication Port to instruct the OMR how the marks on the response form are to be translated into ASCII characters which are usable by the application software.

Download Commands are stored in a volatile memory and consequently, they are only active as long as the OMR-1100 is powered up. For this reason it is recommended that the appropriate download strings are included in the application program when required.

The Factory default conversion at power-up is Binary-To -ASCII

The conversion of the response marks to data may utilize any of four (4) internal formats generated by the OMR logic.

They are:

- Hollerith-To-ASCII Conversion
- Binary-To-ASCII Conversion
- Translate Conversion
- Split Translate Conversion

In addition to the four conversion modes a fifth download command called Row Enable is used in conjunction with any of the translation modes to specify which Rows on the response form are to be considered for translation.

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HOLLERITH TO ASCII CONVERSION:

For those not familiar with the Hollerith form, it was used extensively with early computers as the layout defined for the Punched Card. In Hollerith format each vertical column is defined with 12 keypunch or marking areas. The combination of marks or punches can define the standard 128 character ASCII set. The 12 Channels are defined in two categories. The top three channels are called the "Zone" Channels and the bottom 9 channels are the "Field" channels.

The "Zone" channels are used alone or in combination with "Field" channels (multiple marks or punches per column) to represent the alphabetical and special characters. For definition purposes the top "Zone" is called the "12" channel or "&" channel. The second "Zone" channel is the "11" channel or "-" channel and the third "Zone" channel is the "0" channel.

The "Field" channels are 1 to 9 respectively from top to bottom.

Row	Type	Char
12	Zone	&
11	Zone	-
0	Zone	0
1	Field	1
2	Field	2
3	Field	3
4	Field	4
5	Field	5
6	Field	6
7	Field	7
8	Field	8
9	Field	9

When the reader has been set to Hollerith-To-ASCII Mode each column of marks will be converted to its equivalent ASCII character, as the response form is being read, with one character per column being translated.(See Appendix C)

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BINARY ASCII CONVERSION:

Binary coding utilizes two 8 bit bytes per column to determine which channels in a given column have been marked. This conversion mode will transmit the entire card in an image format that will allow each mark in a column to have specific meaning.

This mode is most commonly used for more complex layouts that utilize multiple marks within a single column to represent specific data or in a form design which contains a layout not applicable to any of the other modes.

In Binary Mode every marking location on the document (max 12 by 126) can be utilized to represent specific data. However, the conversion of the mark, from the Binary ASCII bytes, to a specific variable is controlled by the application program.

Examples of how to convert data from a binary transmission are included in Appendix D.

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TRANSLATE CONVERSION

The translate conversion mode generates a single character per column which is specified in a "Mask" and downloaded to the OMR.

This mode allows 12 characters to be designated to represent the mark in its respective column. In addition, a character may be specified to indicate "no mark" in the column and one to specify that more than one mark was made in a column.

The download commands and "Mask" format are outlined in Section 6.

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SPLIT TRANSLATE CONVERSION

Split-Translate conversion mode generates a single character per column for each half of the card. Up to six characters may be defined to represent the right and left sides of the cards. Two additional characters are used to indicate that no mark existed in the column area, and one to specify that more than one mark was made in a column area.

The download commands and "Mask" format are outlined in Section 6.

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COMBINING CONVERSION MODES:

Most response forms require different layouts within the form design in order to communicate the necessary information. To accommodate this situation the four conversion modes may be used in combination to complete the translation of the entire card. It should be noted that each mode may only be used once for each card definition.

If a specific card layout cannot be translated in this way, use the Binary to ASCII mode and translate the marks within the application program. The sample test cards included with the OMR are a good example. The top portion of the card contains a student I.D. number that can be translated using the Translate Conversion Mode with a numeric "mask" for 0 through 9. The body of the form can be translated using Split-Translate Conversion Mode and a "mask" of A through E to process the multiple choice question.

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ROW ENABLE COMMAND:

This mode can be applied to each conversion mode to disable channels that are not going to be used for data. This reduces the risk of double marks in a data string that could be caused by stray marks, doodling, etc. Row Enable is also useful when used to disable a channel or channels use to indicate the question number or other pre-printed information that is not to be translated.

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SECTION 6 CONFIGURATION DOWNLOAD

There are specific download commands to enable each of the various modes, and a specific format for each command.

Every command must start with a Control R (Decimal 18), followed by the command format.

Multiple commands may be transmitted at one time. The entire download command string is then terminated with the End Of Transmission, "E" command.

The computer to OMR communication must be executed only when the card reader is in an "idle state".

i.e. response forms are not being scanned and the card reader transport is not running.

Standard Download Format:

Ctl R | Command String | Ctl R | Command E

When a command download string has been accepted and processed by the OMR, the character "G" followed by a <CR> will be transmitted by the OMR to the Host computer.

If a command in the download string is NOT executable the download string up to and including the Character that caused an error, followed by "?" and a <CR> will be transmitted by the OMR to the Host computer.

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DOWNLOAD COMMANDS

The following is a list of eight (8) download commands accepted by the OMR:

End of transmitted message "E"
Hollerith - to - ASCII mode "H"
Binary (or image) - to - ASCII mode "I"
Load split translate table "L"
Row enable "R"
Split translate mode "S"
Translate mode "T"
Load Translate Table "X"

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COMMAND E

Action:

Command E is defined as the End of Transmission command.

Syntax:

Chr\$(18);"E"

Remarks:

All download strings transmitted to the OMR must be terminated with Command E. The transmission of only a Command E will clear the card reader of all previous commands that set the mode of operation. i.e. disables all the conversions modes (Commands H,I,S & T).

The transmission of only the Command E will NOT clear any of the previously loaded translation tables or masks. i.e. Commands L,R & X.

Example:

The following code will download the OMR on communication port 1, clearing all conversion mode commands.

```
Open COM1:9600,E as #1
Print #1, Chr$(18);"E"
```

OMR Output:

Returns the character "G" followed by a CR for a successful download.

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COMMAND H

Action:

Command H will set the OMR to Hollerith-To-ASCII mode.

Syntax:

Chr\$(18);"H##"

Where ## is the number of columns to be translated(1 to 99). [00] is used to translate all remaining columns up to a total of 126.

Remarks:

[H] is the character accepted by the OMR to translate marks to the appropriate Hollerith character.

An invalid combination of marks, which has no associated character in the Hollerith Table, will return a "?"

Example:

The following command will place the OMR in Hollerith-To-ASCII mode for 25 columns:

Open COM1:9600,E as #1

DwnLd\$=Chr\$(18)+"H25"+Chr\$(18)+"E"

Print #1, DwnLd\$

Output:

Returns the character "G" followed by a CR for a successful download.

When a card is scanned, one ASCII character per column will be returned followed by a CR. The characters returned will be based on the specific channels marked using the 128 character Hollerith table. (Appendix C)

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COMMAND I

Action:

Command I will set the OMR to Binary-To-ASCII mode that will return 2 bytes per column.

Syntax:

```
Open COM1:9600,E as #1
Chr$(18);"I##"
```

Where ## is the number of columns to be translated(1 to 99) . 00 is used to translate all remaining columns up to a total of 126.

Remarks:

"I" is the character accepted by the OMR to translate all the marks in a column into two Bytes. The marks on the right side of the card(Rows 12,11,0,1,2,3) will be contained in the first byte. The marks on the left side of the card (Row 4,5,6,7,8,9) will be contained in the second byte.

Example:

The following command will place the OMR in Binary-To- ASCII mode for ALL columns:

```
COM1:9600,E As #1
DwnLd$=Chr$(18)+"I00"+Chr$(18)+"E"
Print #1, DwnLd$
```

Output:

A mark is reflected as a 1 bit in the byte and a 0 bit is no mark. The structure of each byte is made up of 8 bits. The right hand bit is the LSB (least significant bit) and the 7th bit is the MSB (most significant bit).

The 8th bit is the parity bit. The 6th bit is forced to a 1 so that a blank column will have a specific value of 32.

Returns the character "G" followed by a CR for a successful download.

When a card is scanned, two bytes per column will be returned followed by a CR. The characters returned will be in the range of 32(Blank) to 127(all 12 channels marked).

The bits are valued from 1 to 64 from the LSB to MSB as follows;

Byte 1:

Row	Value
12	1 LSB
11	2
0	4
1	8
2	16
	32 Always On
3	64 MSB

Byte 2:

Row	Value
4	1 LSB
5	2

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6	4
7	8
8	16
	32 Always On
9	64 MSB

The 7th bit is always "ON" giving each byte a minimum value of 32. This was done to eliminate transmitting any special or control characters, to the Host computer, which all have values less than 32.

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COMMAND L

Action:

Command L defines the conversion table for the Split Translate mode.

Syntax:

```
Chr$(18);"L#####"
```

Where ##### are the six (6) characters to be used in the translation of the marks, plus one (1) for missing mark and one (1) for double mark. The characters are to be used for both sides of the "split" response form. Any of the ASCII characters from 33 to 127 may be used.

Remarks:

"L" is the character accepted by the OMR to set up a conversion table or "mask" to identify which characters are to be used in conjunction with Command S (Split Translate).

The first argument is the character to be used to indicate "no" mark, the next 6 arguments are the characters for the 6 data channels, and the 8th argument is the character to be used for double marks within the 6 data channels. Reference the table below:

Arg. #	Channel
1	NM
2	12 & 4
3	11 & 5
4	0 & 6
5	1 & 7
6	2 & 8
7	3 & 9
8	DM

NM = No Mark DM = Double Mark

Example:

The following basic code will download a split translate conversion table for a multiple choice card, using "*" for no mark and "#" for multiple mark. The character "x" is used to identify unused channels. In this example channels 3 and 9 are used for the question number and will be disabled with Command R.

```
OPEN COM1:9600,E as #1
E$ = Chr$(18)
DwnLd$=E$+"L*EDCBx#"+E$+"E"
Print #1, DwnLd$
```

Output:

Returns the character "G" followed by a CR for a successful download.

When a card is scanned, two ASCII characters per column will be returned followed by a CR.

The characters returned will be based on the relationship of the data mark channel relative to the conversion table. i.e. a marks in channel 8 on the lower or left hand side of the split card would return an "A". A mark in channel 0 in the upper or right hand side of the split card would return a "C", based on the conversion table used in the example.

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COMMAND R

Action:

Command R is defined as the Channel select command.

This command disables or enables specific channels. Disabled channels are "masked" out, therefore marks in a disabled channel are not translated.

Syntax:

Chr\$(18);"R####"

Where #### each of the four arguments are made up of a summed Octal value of three (3) channels each.

Remarks:

Each argument is determined by adding the three values associated with the 4 groups of channels.

The first argument is comprised of channels 9,8 & 7. The second group uses channels 6,5 & 4. The third grouping includes channels 3,2 & 1. The last group is made up of channels 0,11 & 12.

Each argument can have a value of 0 to 7, depending on which channels are to be enabled or disabled. An enable channel is shown as a 1 and disabled channels a 0 (Zero). Channels are assigned a value of 1, 2, or 4.(Octal Coding) from right to left, if they are to be enabled. If they are to be disabled use a value of Zero (0). To determine the value of the argument, add the 3 (three) values together.

The following table illustrates the possible combinations to determine the argument values for each group of channels:

Octal	421	421	421	421	421	421	421	421
Group	000	001	010	011	100	101	110	111
Value	0	1	2	3	4	5	6	7

1 = Enable Channel 0 = Disable Channel

Example:

The following Basic Code will enable channels 12,11,0,1,3,5,7 & 8

Arg. #	#1	#2	#3	#4
Channel	9,8,7	6,5,4	3,2,1	0,11,12
Enable/Disable	#2 0 1 1	0 1 0	1 0 1	1 1 1
Value	3	2	5	7

```
OPEN COM1:9600,E as #1
DwnLd$=Chr$(18)+"R3257"+Chr$(18)+"E"
Print #1, DwnLd$
```

Output:

Returns the character "G" followed by a CR for a successful download.

This Command does not affect the output data string.

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COMMAND S

Action:

Command S will set the OMR to Split Translate Conversion mode.

Syntax:

Chr\$(18);"S##"

Where ## is the number of continuous columns to be translated (1 to 99) . 00 is used to translate all remaining columns up to a total of 126.

Remarks:

"S" is the character accepted by the OMR to translate marks using a split translate layout. This requires that the portion of the response form to be translated in this mode be designed with two columns of up to 6 responses. This layout has the effect of splitting the card into two halves laid out lengthwise of the document.

Example:

The example opens COM1: for output and downloads the Split Translate Conversion command for 25 columns and a Split Translate conversion table. (Command "L")

Open COM1:9600,E as #1

E\$ = Chr\$(18)

E\$+"S25"+E\$+"L*EDCBAX#"+E\$+"E"

Print #1, ChrOut\$

Output:

Returns the character "G" followed by a <CR> for a successful download.

When a card is scanned two ASCII characters per column will be returned followed by a CR. The characters returned will be based on the relationship of the data mark to the character associated with that channel in the conversion table. (i.e. marks in channels 8 and 0 would return an "A" and "C" respectively using a typical multiple choice response form such as the Chatsworth 05-SD or 06-SD cards)

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COMMAND T

Action:

Command "T" will set the OMR to Translate Conversion mode.

Syntax:

Chr\$(18)"T##"

Where ## in the number of continuous columns to translated (1 to 99). 00 is used to translate all remaining columns up to a total of 126.

Remarks:

Command "T" in conjunction with Command "X" gives the host computer the capability of defining the character to be associated with each of the 12 channels on a response form.

Example:

In the example Basic code, the first 9 columns are to translate a student I.D. Number . Where the number goes from 0 to 9, "*" is used to indicate No Mark, and "#" is used to indicate multiple marks. Channels 8 & 9 will be "masked" out with Command R.

Open COM1:9600,E as #1

E\$ = Chr\$(18)

ChrOut\$=E\$+"T09"+E\$+"X*9876543210xx#"+E\$+"E"

Print #1, ChrOut\$

Output:

Returns the character "G" followed by a <CR> for a successful download.

When a card is scanned one ASCII character per column will be returned followed by a <CR>. The characters returned will be based on the relationship of the data mark to the character associated with that channel in the conversion table. (i.e. a mark in channel 6 would return a "1". A mark in channel 12 will return a "9".

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COMMAND X

Action:

Command "X" defines a conversion table for use with Command "T"

Syntax:

Chr\$(18);"X#####"

Where ##### are the twelve (12) characters to be used in the translation of the marks, plus one (1) for missing mark and one (1) for double mark. Any of the ASCII from 32 to 127 may be used.

Remarks:

"X" is the character accepted by the OMR to set up a conversion table to be used in conjunction with Command T (Translate Mode).

The first argument is the character to be used to indicate no mark, the next 12 arguments are the characters for the 12 data channels, and the 14th argument is the character to be used for double marks within the 12 data channels.

The following table illustrates the relationship between the arguments and data channel, or marks:

Arg.	
1	No Mark Character
2	12 channel
3	11 channel
4	0 channel
5	1 channel
6	2 channel
7	3 channel
8	4 channel
9	5 channel
10	6 channel
11	7 channel
12	8 channel
13	9 channel
14	Multiple Mark Character

Example:

The following Basic code will set the OMR to Translate Mode for the first 9 columns of the response form and load a translate table for numeric translations of 0 through 9, "." for no mark and "#" for multiple marks.

The character "x" is used to identify unused channels. In this example channels 9 and 8 are used for the hand written I.D. Number. as illustrated on for the Chatsworth data response form 05-SD

```
OPEN COM1:9600,E as #1
E$ = Chr$(18)
ChrOut$=E$+"T09"+E$+"X*9876543210xx#"+E$+"E"
Print #1, ChrOut$
```

Output:

Returns the character "G" followed by a CR for a successful download.

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When a card is scanned, one ASCII characters per column will be returned column. When the entire response form has been processed the data string will be terminated with a CR.

The characters returned will be based on the relationship of the data mark channel to the character associated with that specific channel in the translate table.

*

COMBINING MODES

Most forms will require more than one command to complete the translation of all the marks on the card to ASCII data. The following command sequences are related to specific response forms that are identified as follows:

Form 1:

Form Number 05-SD - a 100 question multiple choice response form with a 9 column Student I.D. area.

```
OPEN COM1:9600,E as #1
E$ = Chr$(18)
DL1$ = E$+"R3777"
DL2$ = E$+"T09"
DL3$ = E$+"X*9876543210xx#"
DL4$ = E$+"R3737"
DL5$ = E$+"S00"
DL6$ = E$+"L*EDCBAX#"
DL7$ = E$+"E"
DwnLd$=DL1$+DL2$+DL3$+DL4$+DL5$+DL6$+DL7$
Print #1, DwnLd$
```

Form 2:

Form Number H55070-0 is a 35 question response form with a 7 Column I.D. The response area consists of multiple choice questions (A-E) double spaced across the card. Two different translation modes are shown in the examples.

Example 1 Using Binary To ASCII Conversion

```
Open COM1:9600,E as #1
E$ = Chr$(18)
DL1$ = E$+"R1777"
DL2$ = E$+"T07"
DL3$ = E$+"X*9876543210xx#"
DL4$ = E$+"R1252"
DL5$ = E$+"I00"
DL6$ = E$+"E"
ChrOut$=DL1$+DL2$+DL3$+DL4$+DL5$+DL6$
Print #1, ChrOut$
```

Example 2 Using Translate Mode

```
Open COM1:9600,E as #1
E$ = Chr$(18)
DL1$ = E$+"R1777"
DL2$ = E$+"T07"
DL3$ = E$+"X*9876543210xx#"
DL4$ = E$+"R1252"
DL5$ = E$+"T00"
```

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DL6\$ = E\$ + "X*xExDxCxBxAxx#"

DL7\$ = E\$ + "E"

DwnLd\$=DL1\$+DL2\$+DL3\$+DL4\$+DL5\$+DL6\$+DL7\$

Print #1, DwnLd\$

*

Section 7 OMR-1100 EXTENDED COMMANDS

In addition to the Configuration Download Commands there are a series of Extended Commands which are utilized in support of Configuration Options.

Units shipped from the factory will not have any of the Options enabled unless they were specified on the Purchase or Sales Orders.

Since the OMR-1100 Configuration Options are set prior to shipment of the unit they are not downloadable with application software. However, they can be modified, after shipment, with support from our Customer Service Department.

The Configuration Options are stored in non-volatile memory. This means that once a configuration option is set (enabled) it will remain active until it is disabled, even if when power is turned off to the OMR-1100.

The Extended Command Set Includes support for the following options:

- Document Flow Control
- Document Exit Control
- Timing Track Selection
- Data Flow Control
- Data Mark Location
- Parity Format
- Retransmit
- Status
- Version

To implement an Extended Command it is necessary to open the Communications port and then transmit the proper data string to the OMR-1100 to execute the dynamic features of the various options. There are three of the options that do not have dynamic commands; they are Data Flow Control, Data Mark Location and Parity Format.

Examples of specific syntax for Extended Command dynamics will be found in this section along with definitions of the Data Flow Control, Data Mark Location and Parity options, which are not dynamic.

Required Download format:

Ctrl R | ESC | Command [Argument] | Ctrl R | Command E

*

DOCUMENT FEED CONTROL

Action:

Place control of the OMR transport under the control of the Application Software, allowing the transport to be dynamically enabled/disabled.

Syntax:

ESC#

Where # is defined as follows:

D = Disable

E = Enable

Remarks:

Under normal circumstances a simple operator prompt to insert a response form is adequate to control the timing of the feeding response forms. However, in some applications, especially those with extensive processing between documents, it may be desirable to disable the transport while the application software is processing the data string.

If the Document Feed Control option has been set at the factory the ability to feed a response form will be under control of the application program. Upon power up of the unit the motor transport will be disabled.

Consequently, if the operator inserts a response form into the OMR and the application software has not enabled the transport, the OMR will not accent the document. Conversely, when the enable command is sent the transport will remain enabled until a disable command is sent, or a document is processed.

Example:

The following example would enable the OMR Transport

```
Open COM1:9600,E as #1
```

```
ChrOut$=Chr$(18)+Chr$(27)+"E"+Chr$(18)+"E"
```

```
Print #1, ChrOut$
```

*

DOCUMENT EXIT CONTROL:

Action:

Controls the exiting of the response form after the data has been processed, by exiting the form either out the rear of the OMR or returning it to the operator.

Syntax:

ESC##

Where [##] is defined as follows:

ER = Exit Rear

EF = Exit Front

Remarks:

Under the control of the Application Program the response form is transported past the read head and then held in the reader. After the data has been processed the OMR is sent the Exit Command that will continue the Response form through the OMR and out the back or return the document through the entrance throat to the operator.

NOTE: The OMR logic will not accept additional commands while the processor is busy. Therefore, it is necessary to make sure all the data has been received by the Host computer before the eject command is sent.

Example:

The following example would eject the document out the front of the reader.

Open COM1:9600,E as #1

ChrOut\$=Chr\$(18)+Chr\$(27)+"EF"+Chr\$(18)+"E"

Print #1, ChrOut\$

*

DATA FLOW CONTROL

Action:

Utilizes RTS/CTS or Xon/Xoff to control the flow of data from the OMR to the Application Program.

Syntax:

None Required

Remarks:

Data Flow Control has no dynamic command, but is a configuration option only.

This feature is normally used only in networked systems or when the OMR is mounted in a terminal where the host processor is in a multi-tasking environment and must respond to service interrupts. Using this feature allows the application program to receive data from the OMR only when the host is not busy with other functions.

When using the RTS/CTS feature, control is executed by controlling the CTS line from the Host and the RTS line from the OMR. When CTS is held low the OMR will not transmit data or transport a response form. When CTS is held high the OMR will transport and transmit normally.

The OMR will hold RTS low while transporting a document or transmitting data.

NOTE: With the Flow Control option enabled and when using the RTS/CTS feature it is necessary to replace the Chatsworth Data Cable with a standard Null Modem cable.

When using the XOn/XOff feature the OMR will respond normally to the transporting and transmitting when it has received a XOn, and will continue processing documents until a XOff is received. Similar to RTS/CTS the OMR will not accept a document or transmit data when in the Xoff state.

The standard Chatsworth Data cable supplied with the OMR will support the XOn/Xoff feature.

*

TIMING TRACK SELECTION

Action:

Selects the Timing Track Channel

Syntax:

ESC S#

Where [#] is defined as follows;

T = Top timing (Right side as entering OMR)

B = Bottom timing (Left side as entering OMR)

Remarks:

In some applications the location of the Timing Track on the Response Form is critical to the application.

This option allows the OMR to look for a left hand (Bottom) or right hand (Top) timing track or it is possible to look for a timing track on either the left or right edge of the response form. However it should be noted that a single response form may have only ONE timing track.

If "Either" Top timing or Bottom timing was specified, in the factory configuration, for this option then no dynamic command is required since the logic will automatically look for a top or bottom timing track.

Example:

The following example will look for the timing track at the top of the response document.

Open COM1:9600,E as #1

ChrOut\$=Chr\$(18)+Chr\$(27)+"ST"+Chr\$(18)+"E"

Print #1, ChrOut\$

*

DATA MARK ALIGNMENT

Action:

Specifies the location of the data mark in relationship to the timing track. as either "In-Line" or "Off-Set"

Syntax:

None Required

Remarks:

Data Mark Location has no dynamic command, but is a configuration option only.

The location of the Data Mark area with relationship to the timing track has two possible selections.

"In-Line" timing specifies that the leading and trailing edges of the timing mark are within the mark area defined on the Response Form.

"Off-Set" timing specifies that the data mark area lies between the trailing edge of the lead timing mark and the leading edge of the following timing mark.

*

PARITY FORMAT

Action:

Defines the parity format for the data string to be output from the OMR-1100

Syntax:

None Required

Remarks:

Parity Format has no dynamic command, but is a configuration option only.

The default parity format is Even, 7 Bits and 1 Stop Bit. This format will work on all standard P.C.'s and most controllers. However, if a specific serial interface or policy requires other than the default, this option will allow the parity format to be modified.

This option allows the parity to be set to Odd/Even/None with 7 data bits or none with 8 data bits.

The Stop bit is always set at 1

*

RETRANSMIT

Action:

Retransmit the Data Buffer prior to processing the next document.

Syntax:

ESC R

Remarks:

The normal operation of the OMR is to read the document and transmit the data in the specified format. However, under conditions it may be necessary to retransmit the data.

Example:

```
Open COM1:9600,E as #1
ChrOut$=Chr$(18)+Chr$(27)+"R"+Chr$(18)+"E"
Print #1, ChrOut$
```

*

OMR STATUS

Action:

Retrieve the "Status" byte from the OMR-1100

Syntax:

ESC S

Remarks:

The OMR-1100 logic maintains a "Status" byte. Each bit of the byte represents a specific condition of the unit. The definition of each bit is as follows:

Bit 1 = Document at throat

Bit 2 = Document Jam

Bit 3 = Power On Self Test Failure

Bit 4 = Not Used

Bit 5 = Transport Cleared

Bit 6 = Transport Enabled

Bit 7 = Q.C. Test Mode

Bit 8 = Parity Bit

Example:

Open COM1:9600,E as #1

ChrOut\$=Chr\$(18)+Chr\$(27)+"S"+Chr\$(18)+"E"

Print #1, ChrOut\$

Line Input #1, C\$

Locate 10,5:Print "OMR Status ";ASC(C\$)

End

*

VERSION NUMBER

Action:

Transmit the Version Number of the EPROM and date information relative to the current program resident in the OMR-1100

Syntax:

ESC V

Remarks:

The Version Command transmits two characters strings. The first string contains the Model Number of the Unit followed by a space, then a dash, and another space, and the word "Version" and the version number.

[OMR-1100 - Version X.XX]CR

The second string contains the date and time the firmware was generated. The format is as follows;

Day Of Week|Month|Day Of Month|Year|Hour:Min.

[WedNov23199416:23]CR

This information may be helpful when in communications with Chatsworth Customer Service, while diagnosing a problem that may have occurred during set-up or with the operation of the OMR.

The Version Command may also be useful in an application that requires a specific version of the OMR firmware.

Example:

```
Open COM1:9600,E as #1
ChrOut$=Chr$(18)+Chr$(27)+"V"+Chr$(18)+"E"
Print #1, ChrOut$
Line Input #1,C$
Locate 10,5:Print "Version Line 1";C$
Line Input #1,C$
Locate 12,5:Print "Version Line 2";C$
End
```

*

SECTION 8 UTILITY PROGRAMS

The distribution disk supplied with your OMR-1100 contains four (4) utility programs that are supplied to support the use of the OMR-1100. Each of the four utility programs are designed to support specific areas relative to the use of the OMR. The utility programs are:

OMRCfg.EXE	Configuration Update Program
OMRDemo.EXE	Card Image Display
OMRDetct.EXE	Detect OMR Com: Port, including baud rate and cable information.
OMRDnld.EXE	OMR Universal Driver

*

OMRCFG.EXE

Action:

OMRCfg (OMR Configuration Program) is designed to modify the default factory setting for Baud Rate, and Translation Mode. The factory default is 9600 Baud and Binary Translation

Syntax:

OMRCfg [/X/X]

Remarks:

The "Config" program is designed to execute using the OMR factory defaults and COM1. However, if your OMR is no longer set at the factory defaults or the OMR is attached to other than COM1:, use the arguments /X to temporarily modify the program as follows;

/PX where X is a 1 for Com1: or 2 for Com2:

/BX where X is the current baud rate of the OMR (300 -38400)

/Q where Q will cause the OMRCFG.CFG file to download to the OMR.

/? where ? will cause the program to display the above options.

The "Config" program will display a Main Menu as follows;

F1: Help
F5: Restore Factory Defaults
F7: Configure OMR-1100
F10: Save and Exit
EXC: Abandon Changes and Exit

When F7 is selected a sub-menu will display as follows;

ComPort = COM1
Baudrate = 9600
Translation = Binary

Use the up and down arrows to move through the list and the left and right arrows to toggle the default. The following options are available:

ComPort - COM1/COM2
BaudRate: 300 to 38400
Translation: Binary/Hollerith/True Data

Example

The following example will execute the program with the OMR-1100 attached to the COM2: serial communications port.

C:>[CD\OMRUtil] [Enter]
C:>OMRUTIL[OMRCfg /P2] [Enter]

*

OMRDEMO.EXE

Action:

Displays in graphic form the marks as determined by the OMR from any response document.

Syntax:

OMRDemo

Remarks:

OMRDemo is an excellent program to use during checkout of the OMR before beginning the use of an application program or the development of software. This program is also very useful to examine response forms to determine which marks on the document the OMR is detecting as a valid mark.

You may be asked to execute this program when discussing possible problems with the Chatsworth Data Customer Service representative.

Example:

To execute the OMRDemo program, log into the directory which contains the utility program and execute the program by entering the program name.

```
C:>[CD\OMRUtil] [Enter]
```

```
C:>OMRUTIL[OMRDemo] [Enter]
```

Output:

A graphic will display with the Rows of the response form shown horizontally and the Columns display on the vertical axis. Marks are indicated by an "*" and blanks are displayed as ":". The Row and Column locations are aligned on the X/Y axis to assist in locating specific marks on the original document.

*

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OMRDETECT.EXE

Action:

Determines the current configuration of the OMR-1100.

Syntax:

OMRDetect

Remarks:

Utilizes machine code routines to determine the type of configuration the OMR-1100 has been interfaced to and attempts to determine which communications port is being used by the OMR. Then determine the baud rate the OMR is currently operating at, and if possible analyse the data cable being used.

Example:

To execute the OMRDetect program log into the directory which contains the utility program and execute the program by entering the program name.

```
C:>[CD\OMRUtil] [Enter]
```

```
C:>OMRUTIL[OMRDetect] [Enter]
```

Output:

A message is output indicating whether an OMR was found, the Com. Port in use, the Baud Rate, and the cable type, if it can be determined.

*

OMRDnLD.EXE

Action:

Download interface program

Syntax:

OMRDnld

Remarks:

Utilizes command line subroutines to allow a Systems Analyst or Programmer to play "what if" using the various download commands and subsequent displays to determine the format and uses of the various Configuration Download Commands (Section 6) as they would apply to a specific response form.

Example:

To execute the OMRDnld program log into the directory that contains the utility program and execute the program by entering the program name.

```
C:>[CD\OMRUtil] [Enter]
```

```
C:>OMRUTIL[OMRDnld] [Enter]
```

Output:

A visual display of the effect of utilizing the various Configuration Download commands. Error messages and operator prompts assist in the execution of the program.

*

APPENDIX A DOCUMENT SPECIFICATIONS

General:

This specification describes the cards, which can be scanned, on the Chatsworth Data Corporation Optical Mark Sense Card Readers.

Length:

The length of the card should not be less than 3 inches and from a handling standpoint should not exceed 14 inches.

Width:

Card width shall be 3.250 inches +0.007 or -0.003. The card readers will transport a card whose width dimensions range for 3.247" to 3.257". The reading accuracy will decrease if the card width falls below the minimum 3.247" dimension.

Thickness:

The thickness of the card shall be 0.0035" through 0.010"

Composition:

The brightness should be between 81% to 85% reflectance with a maximum of 20% recycle content. A typical stock which should meet this specification would be 36Lb OCR, 70Lb Index, 90Lb Index, or 100Lb Tag

Inks:

For OMR-1100 units with a Visible Red LED (Ink Reader) use PMS-Pantone Warm Red or equivalent.

For OMR-1100 units with an Infer Red LED (Pencil Reader) use any of the following:

PMS-112
PMS-113
PMS-114
PMS-130
PMS-155
PMS-172
PMS-185
PMS-191
PMS-198
PMS-218
PMS-252

Grain:

The grain of the paper fibres of the card specified for the OMR-1100 may either be parallel to the length of the document (long grain) or parallel to the width of the document (short grain) as long as the maximum card dimensions are not exceeded.

Curl:

The maximum allowable curl of the card measuring from a flat surface on which a card is laying to the most protruding point of curl on the card is 0.05". The maximum allowable curl dimension applies to when the card is placed both face up or face down.

Corner Cut:

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A corner cut on the card is not required for use with the OMR-1100 reader. If a corner cut is used, the preferred placement would be at the trailing edge of the card, and should be a diagonal cut as illustrated by Figure A-1.

Rounded Corners:

The use of rounded corners is not required for use with the OMR-1100 reader. However, you can enhance the feeding of the cards using rounded corner cards. When used the rounded corner cuts should be applied as illustrated in Figure A-2.

Perforations:

If possible, avoid using perforations on the leading edge or the timing track edge of the card. Using strip or litho perforations is NOT recommended. For the best result, use 12 to 30 tooth blade or wheel. All cuts should be crisp with no perforations embossing.

Timing Marks:

Timing marks may be printed on either edge of the response document. However, the factory default is defined as bottom timing. This would place the timing tack on the left side of the response document as it is entered into the OMR-1100.

The timing marks define the location and the depth of the mark area for each data channel within the response form. This allows for flexibility in forms design as only response areas defined by a timing mark will be scanned for a mark.

With "In-line" timing the depth of the response area is defined from the leading edge of the timing mark to the trailing edge of the same timing mark. With "Off-set" the response area is defined from the trailing edge of one timing mark to the leading edge of the next timing mark.

Individual timing marks may be spaced as required by the design of the form. However, the minimum spacing between timing marks should not be less than 10 to the inch. It is recommended that for hand marked areas that no more than 8 timing marks per inch be used. The most common spacing for hand marked areas is 6 per inch.

Timing mark specifications are shown on Figure B

Laser Compatibility:

It is feasible to imprint on cards using most laser printer. For example you may wish to imprint a student name and number and then encode a student I.D. number in Hollerith code. Thickness of the stock should be tested on your laser, as the feeding of the card stock will vary from printer to printer. It is recommended that you use non-scan laser compatible inks when printing the card stock.

*

APPENDIX B

FACTORY DEFAULTS

Baud Rate:	9600
Parity:	Even
Data Mark Alignment:	In-Line
Timing Track Selection:	Bottom (Left Side)
Data Mark Conversion:	Binary ASCII
Row Enable:	All Rows Enabled
Document Feed Control:	Disabled
Document Exit Control:	Feed Through
Data Flow Control:	None

APPENDIX C HOLLERITH TO ASCII CONVERSION

ASCII		Hollerith	
Dec	Oct	Char	

0	000	Nul	9-8-1-0-12
1	001	SOH	1-12
2	002	STX	9-2-12
3	003	ETX	9-3-12
4	004	EOT	9-7
5	005	ENQ	9-8-5-0
6	006	ACK	9-8-6-0
7	007	BEI	9-8-7-0
8	010	BS	9-6-11
9	011	HT	9-5-12
10	012	LF	9-5-0
11	013	VT	9-8-3-12
12	014	FF	9-8-4-12
13	015	CR	9-8-5-12
14	016	SO	9-8-6-12
15	017	SI	9-8-7-12
16	020	DLE	9-8-1-11-12
17	021	DC1	9-1-11
18	022	DC2	9-2-11
19	023	DC3	9-3-11
20	024	DC4	9-8-4
21	025	NAK	9-8-5
22	026	SYN	9-2
23	027	ETB	9-6-0
24	030	CAN	9-8-11
25	031	EM	9-8-1-11
26	032	SUB	9-8-7
27	033	ESC	9-7-0
28	034	FS	9-8-4-11
29	035	GS	9-8-5-11
30	036	RS	9-8-6-11
31	037	US	9-8-7-11
32	040	SP	(Blank)
33	041	!	8-7-12
34	042	@	8-7
35	043	#	8-3
36	044	\$	8-3-11
37	045	%	8-4-0
38	046	^	12
39	047	'	8-5
40	050	(	8-5-12
41	051	)	8-5-11
42	052	*	8-3-0
43	053	+	8-4-11
44	054	,	8-3-0
45	055	-	11
46	056	.	8-3-12

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47	057	/	1-0
48	060	0	0
49	061	1	1
50	062	2	2
51	063	3	3
52	064	4	4
53	065	5	5
54	066	6	6
55	067	7	7
56	070	8	8
57	071	9	9
58	072	:	8-2
59	073	;	8-6-11
60	074	<	8-4-12
61	075	=	8-6
62	076	>	8-6-0
63	077	?	8-7-0
64	100	@	8-4
65	101	A	1-12
66	102	B	2-12
67	103	C	3-12
68	104	D	4-12
69	105	E	5-12
70	106	F	6-12
71	107	G	7-12
72	110	H	8-12
73	111	I	9-12
74	112	J	1-11
75	113	K	2-11
76	114	L	3-11
77	115	M	4-11
78	116	N	5-11
79	117	O	6-11
80	118	P	7-11
81	121	Q	8-11
82	122	R	9-11
83	123	S	2-0
84	124	T	3-0
85	125	U	4-0
86	126	V	5-0
87	127	W	6-0
88	130	X	7-0
89	131	Y	8-0
90	132	Z	9-0
91	133	[	8-2-0
92	134	\	8-2-11
93	135	]	8-2-11
94	136	^	8-7-11
95	137	~	8-5-0
96	140	`	8-1
97	141	a	1-0-12
98	142	b	2-0-12
99	143	c	3-0-12
100	144	d	4-0-12
101	145	e	5-0-12

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102	146	f	6-0-12
103	147	g	7-0-12
104	150	h	8-0-12
105	151	i	9-0-12
106	152	j	1-11-12
107	153	k	2-11-12
108	154	l	3-11-12
109	155	m	4-11-12
110	156	n	5-11-12
111	157	o	6-11-12
112	160	p	7-11-12
113	161	q	8-11-12
114	162	r	9-11-12
115	163	s	2-0-11
116	164	t	3-0-11
117	165	u	4-0-11
118	166	v	5-0-11
119	167	w	6-0-11
120	170	x	7-0-11
121	171	y	8-0-11
122	172	z	9-0-11
123	173	{	0-12
124	174		11-12
125	175	}	0-11
126	176	~	1-0-11
127	177	DEL	9-7-12